The Al-Er-Mg Ternary System Part II: Thermodynamic Modeling

G. Cacciamani, A. Saccone, S. De Negri, and R. Ferro

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A thermodynamic modeling and optimization of the Al-Er, Er-Mg, and Al-Er-Mg systems has been carried out, the Al-Mg system having already been optimized in the literature. The sublattice model has been used to describe the thermodynamic functions of both the solution phases and the stoichiometric phases present in the systems. In particular, the order/disorder relationship between the couples of phases A2/B2 and A1/L1$_2$ present in the system as bcc-(Er,Mg)/Er(Mg$_{1-x}$Al$_x$) and fcc-(Al)/(Al$_{1-x}$Mg$_x$)$_3$Er, respectively, has been taken into account and thermodynamically modeled. Moreover γ-(Al,Mg) and Er$_5$(Mg$_{1-x}$Al$_x$)$_{24}$ phases, both related to the cI58-αMn (A12) type structure, have been modeled as different sublattice occupations of the same A12 phase. The thermodynamic modeling and the experimental investigation of the phase equilibria (reported in part I of this work) have been carried out by a recursive procedure. As a result, complete descriptions of the solid-solid and solid-liquid phase equilibria have been obtained.

1. Introduction

In part I of this paper [2002Sac], we reported the results of an experimental investigation of the Al-Er-Mg system. This second part concerns the thermodynamic modeling and optimization of the same system.

The combination of experiments and thermodynamic calculations is a powerful strategy when exploring multicomponent systems. Indeed, it was successfully applied to the study of the Al-Mg-Sc system [1998Pis, 1999Cac, 1999Gro]. Also, in this case experiments and thermodynamic calculations were carried on by an iterative procedure. Calculations were used to interpret experimental results and to select new key experiments; experimental results, in turn, were used to refine the thermodynamic description of the phases and to obtain more reliable calculations.

Two of the three binary subsystems, Al-Er and Er-Mg, were optimized in this work, and one of them, Al-Er, required further experimental investigation, as reported in part I [2002Sac]. For the Al-Mg system, the optimization results by Liang et al. [1998Lia] were adopted. In the following, for brevity, shorter and structure-related phase names will be used. See Table 1 for the list of the phases present in the system.

2. Experimental Information Used for the Optimization

2.1 Al-Er System

Phase equilibria presented in the assessment by Gschneidner and Calderwood [1998Gsc], mainly based on the work by Buschow [1965Bus], generally were used. However, the curvature variations shown by the liquidus curve in the 50 to 80 at.% Er composition range were considered quite improbable and suggested a reinvestigation of this portion of the phase diagram. The new results, presented and discussed in part I, were also used. Thermodynamic information useful for the system optimization is very scarce. Colinet and Pasturel [1985Col] and Sommer and Keita [1987Som] determined the enthalpy of formation of Al_2Er by calorimetry and [1983Mer] the heat capacity of the same phase in the 400 to 1200 K temperature range.

2.2 Er-Mg System

The Er-Mg phase equilibria have been determined by Saccone et al. [1992Sac]. Only the solubility of Er in (Mg) at the Mg-rich eutectic temperature had been previously studied by Rokhlin [1978Rok]. The Gibbs energy of formation of ErMg, $ErMg_2$, and Er_5Mg_{24} has been obtained by Pahlman and Smith [1972Pah] from vapor pressure measurements by the Knudsen effusion technique. All these data were used in the optimization.

2.3 Al-Er-Mg System

The isothermal phase equilibria at 400 °C and the liquidus temperatures reported in part I have been used for the optimization. Other subliquidus thermal effects detected by differential thermal analysis (DTA) have been compared to the computation results but were not used for the optimization. No thermodynamic data are known in the literature for this system.

3. Thermodynamic models

The thermodynamic models adopted here for the Al-Er-Mg phases are summarized in Table 1 and are briefly introduced in the following.

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Table 1 Summary of the Sublattice Models Adopted

Phase Name	Structure Type	Sublattice Model (a)	Comment
Liquid		(A1,Er,Mg)	Disordered phase
(Al) or A1	cF4-cu	(Al,Er,Mg)	Disordered phase
bcc-(Er,Mg) or A2	cI2-W	(Al,Er,Mg)	Disordered phase
(Mg) or (Er) or A3	hP2-Mg	(Al,Er,Mg)	Disordered phase
$(Mg_{1-x}Al_x)Er$ or B2	cP2-CsC1	$(Al,Er,Mg)_{0.5}(Al,Er,Mg)_{0.5}$	Ordered Er-Mg binary phase related to A_2, with disordered solution of Al
$(Al_{1-x}Mg_x)_3Er$ or $L1_2$	cP4-$AuCu_3$	$(Al,Er,Mg)_{0.75}(Al,Er,Mg)_{0.25}$	Ordered Al-Er binary phase related to A_1, with disordered solution of Mg
γ-(Al,Mg) or A12-(Al,Mg)	cI58-αMn	$(Er,Mg)_5(Al,Mg)_{12}(Al,Mg)_{12}$	The two binary phases correspond to different occupation states of the same sublattices
$(Mg_{1-x}Al_x)_{24}Er_5$ or A12-(Er,Mg)		$(Er,Mg)_5(Al,Mg)_{12}(Al,Mg)_{12}$	
$(Al_{1-x}Mg_x)_2Er$ or C15	cF24-$MgCu_2$	$(Al,Er,Mg)_2(Al,Er,Mg)_1$	Ordered phase
$(Mg_{1-x}Al_x)_2Er$ or C14	hP12-$MgZn_2$	$(Er,Mg)_2(Er,Mg)_1$	Ordered phase
$(Al_{1-x}Mg_x)Er$ or AlEr	oP16-AlEr	$(Al,Mg)_1(Er)_1$	Ordered phase
$(Al_{1-x}Mg_x)_2Er_3$ or Al_2Er_3	tP20-Al_2Zr_3	$(Al)_2(Er)_3$	Stoichiometric compound
$(Al_{1-x}Mg_x)Er_2$ or $AlEr_2$	oP12-$SiCo_2$	$(Al)_1(Er)_2$	Stoichiometric compound
β-(Al,Mg) or Al_3Mg_2	cF1832-Mg_2Al_3	$(Mg)_{89}(Al)_{140}$	Stoichiometric compound
ε-(Al,Mg)	hR159-$Co_5Cr_2Mo_3$	$(Mg)_{23}(Al)_{30}$	Stoichiometric compound
τ	related to cF24-$MgCu_2$	$(Al)_{0.667}(Er)_{0.1}(Mg)_{0.233}$	Stoichiometric compound

(a) Predominant element in each sublattice, if any, is underlined.

3.1 Elements

The Gibbs energies of the pure elements are referred to the enthalpy of each element in its so-called standard element reference (SER) state (the stable state at 10^5 Pa and 298.15 K).

The Gibbs energy functions of the pure elements in the stable phases (A1 and liquid Al, A3, and liquid Er and Mg) were taken from the PURE database [1996SGT] and are reported in the Appendix. The Gibbs energy functions of the pure elements in metastable structures (reference values for the thermodynamic description of the solid solutions) are discussed separately for each phase in Section 4.

3.2 Phases

The Gibbs energy of the various phases is expressed by the sublattice model [1981Sun]. According to this model, the different atoms forming the phase occupy one or more sublattices, and thermodynamic quantities are expressed as a function of the site fractions. The model is suitable for the description of either stoichiometric compounds and of both disordered and ordered solutions.

Each sublattice contains all the atoms having similar interactions with the surrounding. In general, sublattices may correspond to the Wyckoff special positions in the crystal lattice; however, especially in the case of complex structures, one sublattice may include the atoms contained in two or more Wyckoff special positions, in order to reduce the total number of sublattices and, consequently, the number of parameters to be optimized.

The total Gibbs energy of a phase φ is given by the sum of three contributions.

$$G^{\varphi} = {}^{\mathrm{ref}}G^{\varphi} + {}^{\mathrm{id}}G^{\varphi} + {}^{\mathrm{ex}}G^{\varphi}$$

where the three terms assume different forms according to the kind of sublattice model adopted.

Stoichiometric Compounds. Ordered stoichiometric compounds are represented with as many sublattices as the number of component elements, with only one atom type in each sublattice. In the binary case, the model is $(A)_u(B)_v$, and the three contributions to the Gibbs energy are:

$$\begin{aligned} {}^{\mathrm{ref}}G^{\varphi} &= u/(u+v)\, G_A^{\mathrm{SER}} + v/(u+v)\, G_B^{\mathrm{SER}} + L_{A:B}^{\varphi} \qquad \text{(Eq 1)} \\ {}^{\mathrm{id}}G &= 0 \\ {}^{\mathrm{ex}}G &= 0 \end{aligned}$$

where $L_{A:B}^{\varphi} = \Delta_f G(\varphi)$ is the Gibbs energy of formation of the phase referred to the SER state of the component elements.

This model was adopted for the binary Al_2Er_3, $AlEr_2$, β, and ε phases and for the ternary τ phase, (the model is the obvious generalization of the binary one). In all cases the L interaction functions were assumed to be:

$$L = a + b\,T \qquad \text{(Eq 2)}$$

where a and b are temperature- and composition-independent parameters to be optimized.

Disordered Substitutional Solutions. Disordered substitutional solutions are represented by only one sublattice where all the atoms mix together. In the binary case (A,B) the three contributions to the Gibbs energy are:

$$\begin{aligned} {}^{\mathrm{ref}}G^{\varphi} &= x_{\mathrm{A}}\, G_A^{\varphi} + x_{\mathrm{B}}\, G_B^{\varphi} \qquad \text{(Eq 3)} \\ {}^{\mathrm{id}}G^{\varphi} &= R\,T\,[x_{\mathrm{A}} \ln(x_{\mathrm{A}}) + x_{\mathrm{B}} \ln(x_{\mathrm{B}})] \\ {}^{\mathrm{ex2}}G^{\varphi} &= x_{\mathrm{A}}\, x_{\mathrm{B}}\, \Sigma_v\, {}^{v}L_{\mathrm{A,B}}^{\varphi}\, (x_{\mathrm{A}} - x_{\mathrm{B}})^{v} \end{aligned}$$

where G_A^{φ} and G_B^{φ} are the Gibbs energies of the pure elements in the φ structure and the L functions have the same form as before. In the ternary case, binary excess terms are combined according to the Muggianu et al. extrapolation formula [1975Mug], and a ternary excess term is added (only the 0th order term was used):

$$^{ex3}G^{\varphi} = x_A\, x_B\, x_C\, L_{A,B,C}^{\varphi}$$

where the L function still has the form shown in Eq (2). This model was adopted for the liquid and for the A1, A2, and A3 solid solutions.

Ordered Solid Solutions. Ordered solid solutions are represented by two or more sublattice models where the component elements are allowed to mix in one or more sublattices. A binary phase of this kind may be represented by the model $(\underline{A},B)_u(A,\underline{B})_v$, the first sublattice being mainly occupied by A and the second by B. If y_A', y_A'', y_B' and y_B'' are the site fractions of the elements A and B in the first (′) and second (″) sublattice, respectively, the fully ordered state corresponds to the conditions $y_A' = 1$, $y_A'' = 0$, $y_B' = 0$, and $y_B'' = 1$; the fully disordered state corresponds to the condition $x_B = y_B' = y_B''$ (same composition in each sublattice and in the overall phase).

The contributions to the Gibbs energy are:

$$^{ref}G^{\varphi} = y_A' y_A''\, G_{A:A}^{\varphi} + y_A' y_B''\, G_{A:B}^{\varphi} + y_B' y_A''\, G_{B:A}^{\varphi} + y_B' y_B''\, G_{B:B}^{\varphi} \qquad \text{(Eq 4)}$$

$$^{id}G^{\varphi} = R\,T[u/(u+v)(y_A' \ln(y_A') + y_B' \ln(y_B')) + v/(u+v)\,(y_A'' \ln(y_A'') + y_B'' \ln(y_B''))]$$

$$\begin{aligned} ^{ex}G^{\varphi} = {} & y_A' y_B' y_A''\, \Sigma_v\, {}^vL_{A,B:A}^{\varphi}\,(y_A' - y_B')^v \\ & + y_A' y_B' y_B''\, \Sigma_v\, {}^vL_{A,B:B}^{\varphi}\,(y_A' - y_B')^v \\ & + y_A' y_A'' y_B''\, \Sigma_v\, {}^vL_{A:A,B}^{\varphi}\,(y_A'' - y_B'')^v \\ & + y_B' y_A'' y_B''\, \Sigma_v\, {}^vL_{B:A,B}^{\varphi}\,(y_A'' - y_B'')^v \\ & + y_A' y_B' y_A'' y_B''\, \Sigma_v\, {}^vL_{A,B:A,B}^{\varphi}\,(y_A' - y_B')^v\,(y_A'' - y_B'')^v \end{aligned}$$

It may be emphasized that when more atoms mix in a sublattice (see Eq 3) $^{ref}G^{\varphi}$ requires the Gibbs energy of the pure elements in the φ phase to be known even if it is not a stable phase for the element (consider, as an example, Er in the A1 or A2 phases). Similarly, when more sublattices are used and more atoms occupy each sublattice, terms of the type $G_{A:B}^{\varphi}$ appear in $^{ref}G^{\varphi}$ (see (4)). They correspond to the Gibbs energies of formation of hypothetical stoichiometric compounds, most of which are metastable in the system. Nevertheless they have to be evaluated in some way, together with the $^vL^{\varphi}$ parameters which still have the form shown in Eq 2.

Order-Disorder Relations. Phases related by an order-disorder transformation, like A1 (cF4-Cu) and $L1_2$ (cP4-$AuCu_3$), can be described as a single phase, the Gibbs energy of which is expressed by the following equation [1997Ans1].

$$\begin{aligned} G^{\varphi}(y_i', y_i'') &= {}^{dis}G^{\varphi}(x_i) + {}^{ord}G^{\varphi}(y_i', y_i'') \\ &= G^{\psi}(x_i) + G^{\varphi}(y_i', y_i'') - G^{\varphi}(x_i) \end{aligned}$$

where φ and ψ are the related ordered and disordered phases, respectively; $G^{\psi}(x_i)$ and $G^{\varphi}(x_i)$ are calculated according to the Eq 3, and $G^{\varphi}(y_i', y_i'')$ is calculated according to the Eq. 4. In the present system, two couples of phases show such an order-disorder relation: A2 and B2; and A1 and $L1_2$. In both cases, in order to reduce the number of independent parameters to be optimized, a number of constraints among the G or L quantities may be assumed (see [1997Ans] for the A2 and $L1_2$ phases and [1999Dup] for the A2 and B2 phases).

Order-Order Relations. The crystal structures of γ-AlMg and Er_5Mg_{24} binary phases correspond to two different occupation states of the cI58-αMn-type structure, often denoted in literature as the A12 or χ phase. The thermodynamic modeling of this phase has been widely discussed by [1997Ans2], and its conclusions have been adopted here. Then the two phases have been modeled as different sublattice occupations of the same phase here called A12 (Table 1).

4. Optimization

For the Al-Mg system, the thermodynamic optimization by [1998Lia] was adopted.

The binary Al-Er and Er-Mg systems were first approximately optimized by considering all the intermediate phases as stoichiometric. Starting values for the thermodynamic functions of the phases were estimated by interpolation. It is well known, in fact, that most properties of the rare earth (RE) metal alloys show smooth and regular variations as a function of the RE atomic number. As a consequence, lacking values for a specific RE-metal alloy may be reliably estimated by interpolation whcn the data for other RE-metal alloys are available. A first, roughly approximated, calculation of the ternary Al-Er-Mg phase equilibria was then obtained by the ideal combination of the three binary subsystems.

After that, the phase models and the optimizing parameters were recursively refined taking into account at each step the results of new experiments performed on the basis of the indications of the previous calculations. The final thermodynamic description of the complete ternary system is herein presented, phase by phase, and the optimization results are summarized in the Appendix, where the parameters calculated in this work are underlined.

4.1 Liquid

No thermodynamic data were available on the binary Al-Er and Er-Mg liquid phases. The Gibbs energy was first estimated on the basis of the data available for analogous Al-Ln or Ln-Mg systems and then optimized considering the phase equilibria. Only one ternary excess parameter was used to satisfactorily reproduce the ternary liquid equilibria.

4.2 A1 (cF4-Cu)

This is the reference state structure for Al. No data were available in the PURE database for the metastable A1-Er, it was assumed that

$$G_{Er}^{A1} = 1950 - 0.1\,R\,T + G_{Er}^{SER}$$

This was estimated by assuming $\Delta S^{A1\text{-}A3} = 0.1$ R, which is in agreement with [1988Fer], and by evaluating $\Delta H^{A1\text{-}A3}$ by extrapolation to the pure Er of the A1 + A3 two-phase field in the binary Er-Th system. The very small solid solubility of Er in A1-Al was neglected, and no binary Al-Er excess function was optimized. An Er-Mg binary excess parameter, however, was calculated, even though the A1 phase does not appear in the system, in order to reproduce the experimentally determined solid solubility of Mg in (Al_3Er) with due note that the ordered $L1_2$ phase is related to A1.

4.3 A2 (cI2-W)

None of the three component elements is stable in this structure, but Gibbs energy data are reported in the PURE database for Al and Mg. G_{Er}^{A2} was taken from [2000Nor]. The Al-Er excess function was optimized on the basis of the B2 (the ordered phase related to A2) ternary solubility. Ideal behavior in the Al-Er and Al-Mg systems (where the phase is metastable) was assumed. No ternary interaction parameters were added.

4.4 A3 (hP2-Mg)

This is the reference state structure for Er and Mg. The very small solid solubility of Al in A3-Er was neglected, and no binary Al-Er excess function was optimized. No ternary parameter was needed.

4.5 $L1_2$ (cP4-$AuCu_3$)

This phase was modeled, in the binary Al-Er system, as the ordered form of A1. The expression of the binary excess functions was determined according to [1997Ans2]. No ternary parameter was needed to correctly reproduce the small ternary solubility of this phase.

4.6 B2 (cP2-CsCl)

This phase was modeled, in the binary Er-Mg system, as the ordered form of A2, and the expression of the binary Er-Mg excess functions was determined according to [1999Dup]. In order to reproduce the large ternary solubility of this phase, binary Al-Er and ternary excess functions were also optimized.

4.7 A12 (cI58-αMn)

According to the model adopted, this phase is not defined in the whole ternary composition range: in particular, it does not include the pure Al and Er. G_{Mg}^{A12} is given in the PURE database. The other G terms appearing in ${}^{ref}G^{A12}$, which were not already optimized by [1998Lia], and corresponding to the Gibbs energy of formation of hypothetical metastable ternary compounds, were estimated by linear combination of those already determined. This is equivalent to the assumption that, in each sublattice, the enthalpy and entropy of substitution of one atom by another are only dependent on the nature of the two atoms and are independent of the composition, the temperature, and the occupation of the other sublattices.

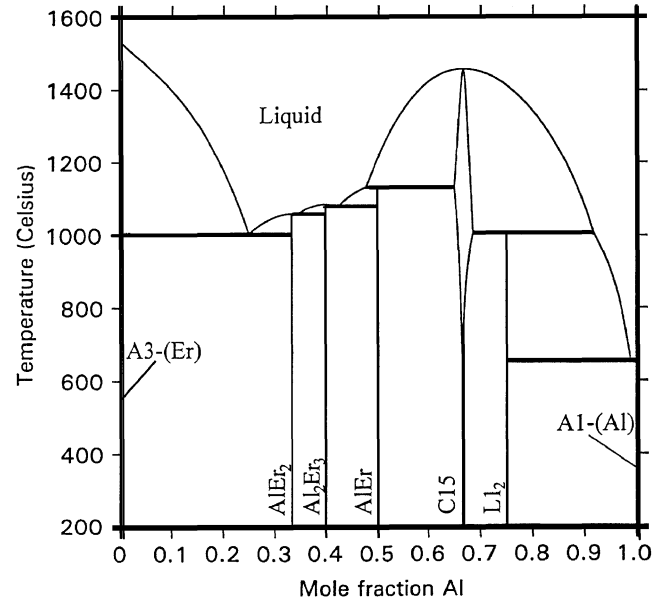

Fig. 1 The computed Al-Er phase diagram

4.8 C14 and C15 (cF24-$MgCu_2$ and hP12-$MgZn_2$)

These two phases were modeled according to the general assumptions usually adopted for the Laves-type phases. C15 is stable in the Al-Er system as Al_2Er, and C14 is stable in the Er-Mg system as $ErMg_2$. Being structures closely related to each other, it was assumed that in each binary system the excess Gibbs energy of the metastable phase (C15 in Er-Mg and C14 in Al-Er) is not much different from that of the stable one. By this assumption, the ternary solubility of C15 was correctly described without the addition of ternary interaction parameters. The C14 phase, however, was indicated to have a negligible ternary solubility (in agreement with the experimental information) and was then modeled as a binary Er-Mg phase in the final optimization.

All the remaining phases, Al_2Er_3, $AlEr_2$, β-(Al,Mg), ε-(Al,Mg), and τ, were described as stoichiometric, even if very small or uncertain solubility ranges were experimentally detected.

5. Results and Discussion

The Gibbs energy functions for all the phases in the Al-Er-Mg system are summarized in the Appendix, where the underlined parameters are those optimized in this work. The calculated phase equilibria are presented and briefly discussed in the following.

5.1 Al-Er System

The optimized Al-Er phase diagram is shown in Fig. 1. It may be compared to the assessed version presented by [1988Gsc] taking into account also the new results reported in part I of this paper. Owing to the new experiments, the equilibria in the 50 to 80 at.% Er are still defined with some uncertainty, mainly with regard to the invariant tempera-

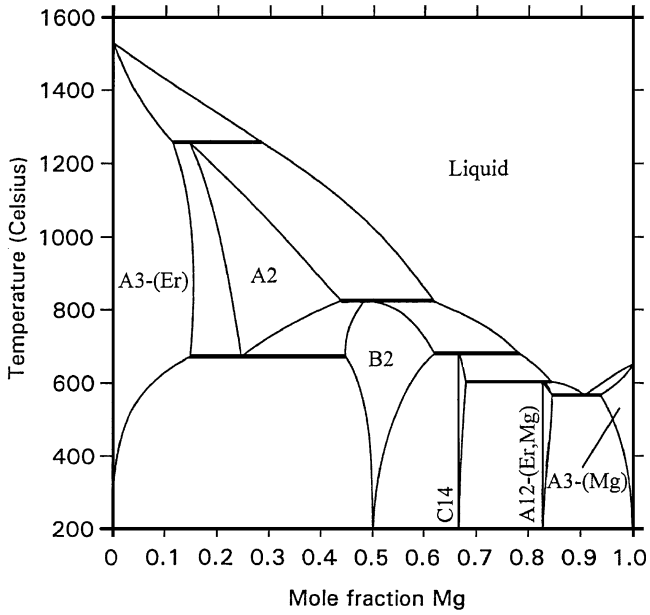

Fig. 2 The computed Er-Mg phase diagram

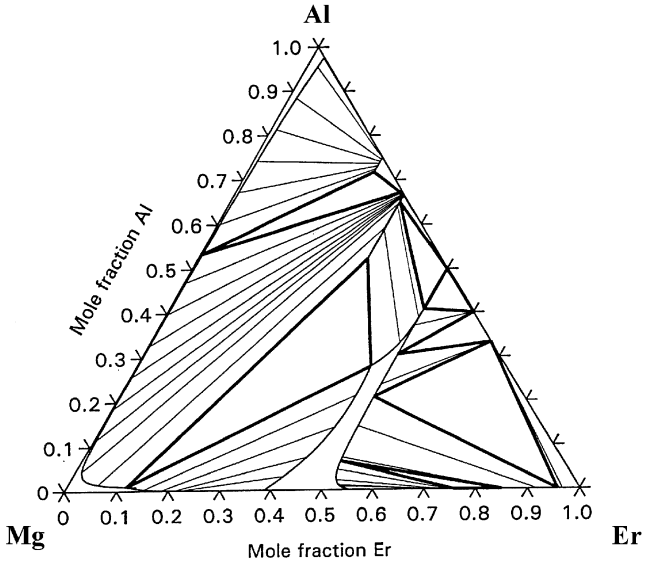

Fig. 4 Al-Er-Mg system: isothermal section computed at 700 °C

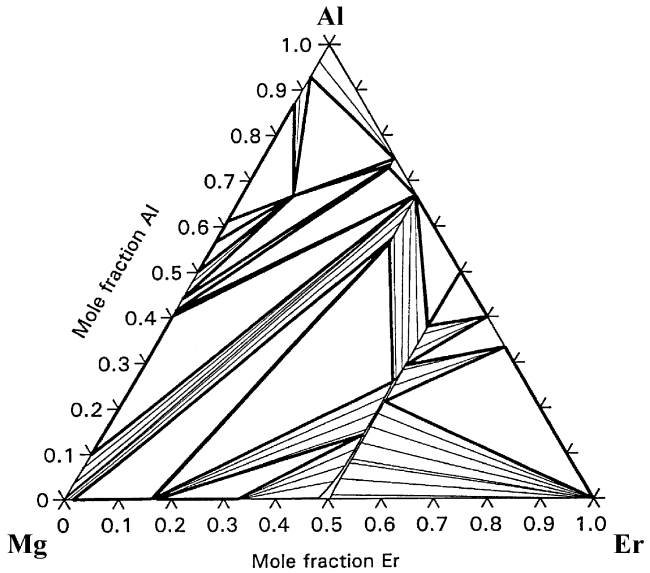

Fig. 3 Al-Er-Mg system: isothermal section computed at 400 °C

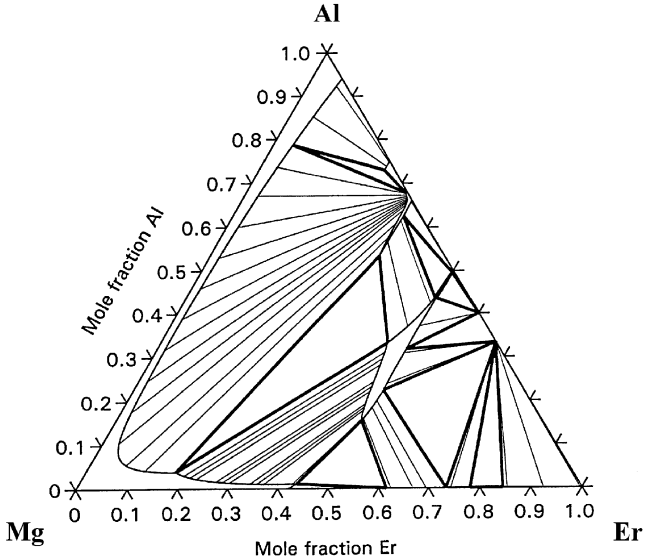

Fig. 5 Al-Er-Mg system: isothermal section computed at 900 °C

tures and the melting behavior of the $AlEr_2$ phase. In the computed phase diagram, the melting behavior of the $AlEr_2$ phase is congruent (even if very close to the eutectic temperature and composition), and the invariant temperatures slightly differ from the experimental ones It was considered unnecessary to introduce more complex thermodynamic functions in order to match more closely such uncertain experimental data.

In the 70 to 100 at.% Al region, the experimental liquidus curve looks depressed with respect to the computed one: in this case, the computed curve is considered more reliable than the experimental one, which presents an unlikely steep trend. Similar unlikely curvature of the experimental liquidus may be observed in several Al-R binary systems; it is possibly due to undercooling effects. The Al_3Er peritectic decomposition temperature is lower than that reported by [1988Gsc] but is in better agreement with the ternary-phase equilibria.

5.2 Er-Mg System

The optimized Er-Mg phase diagram is shown in Fig. 2. It is in good agreement with the experimental version reported by [1992Sac]. The only appreciable difference is the composition of the A2 phase at the eutectoidic equilibrium, which is calculated to be about 5 at.% richer in Er than the experimental determination.

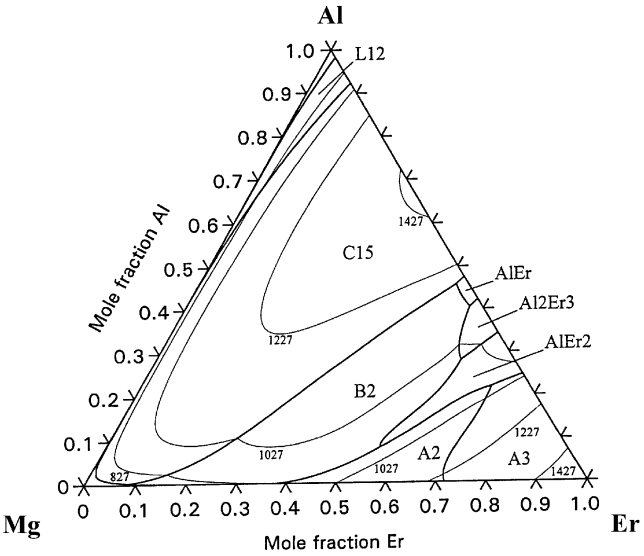

Fig. 6 Al-Er-Mg system: the computed liquidus surface

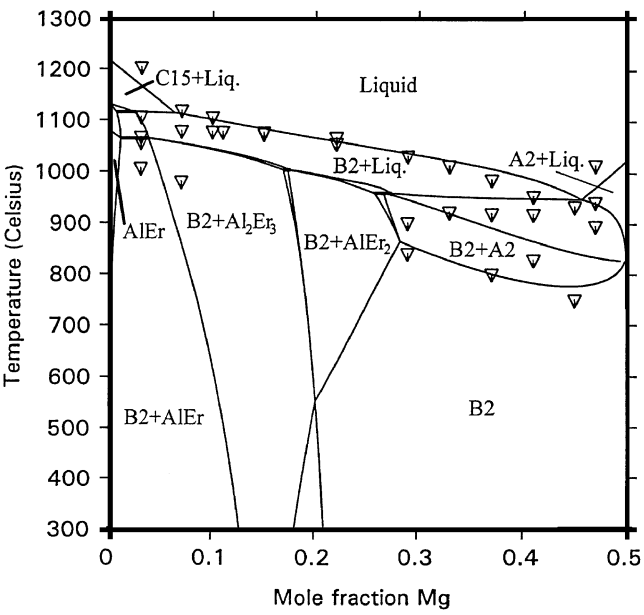

Fig. 7 Al-Er-Mg system: vertical section computed at 50 at.% Er

5.3 Al-Er-Mg System

The optimized Al-Er-Mg isothermal section at 400 °C is shown in Fig. 3. It is in very good agreement with the experimental results described in part I of this paper. Small discrepancies between experimental and calculated equilibrium compositions of the solution phases in a few three-phase equilibria may be due to an insufficient annealing of the samples: in most cases, in fact, the experimentally determined compositions correspond to those stable at higher temperatures. A few more isothermal sections, which were calculated at 700 and 900 °C, are shown in Fig. 4 and 5, respectively.

The projection of the liquidus surface is reported in Fig. 6. The extent of the primary solidification regions for the different phases is in good agreement with the experimental information reported in part I. As expected, considering the high stability of the C15 phase, a large area of the ternary field is dominated by the phase. In this region, the computed liquidus surface often lays at temperatures slightly lower

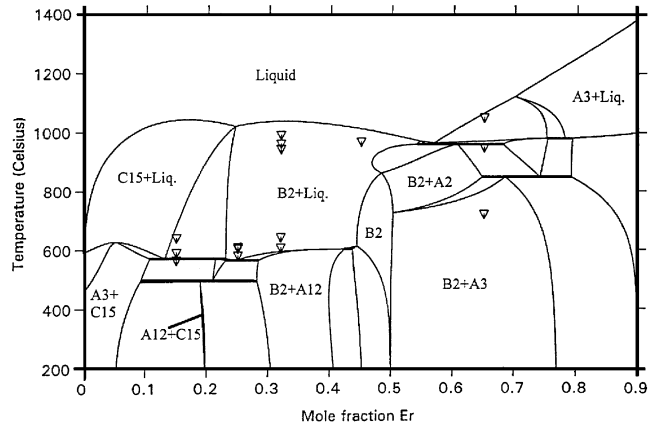

Fig. 8 Al-Er-Mg system: vertical section computed at 10 at.% Al

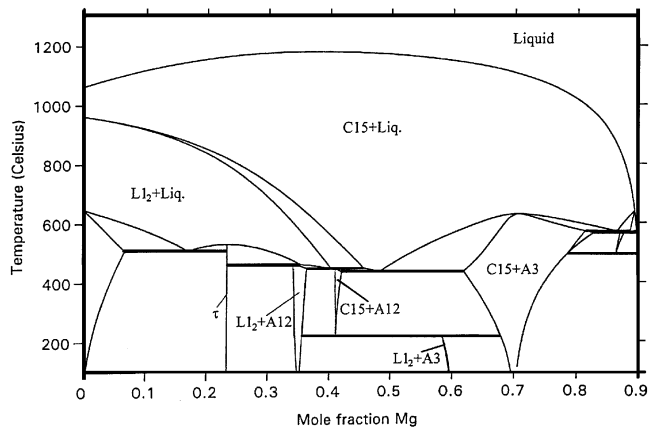

Fig. 9 Al-Er-Mg system: vertical section computed at 10 at.% Er

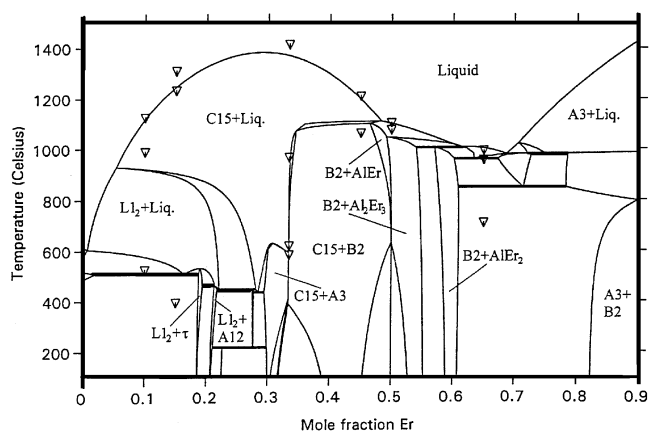

Fig. 10 Al-Er-Mg system: vertical section computed at 10 at.% Mg

Table 2 Calculated Phase Compositions at the Invariant Equilibria in the Al-Er-Mg System

Invariant Equilibrium	Involved Phases	Composition		
		at.% Al	at.% Er	at.% Mg
P1	L	45.7	51.8	2.5
	C15	62.6	35.0	2.4
	AlEr	49.4	50.0	0.6
	B2	48.5	49.1	2.4
U1	L	40.5	57.1	2.4
	AlEr	49.0	50.0	1.0
	Al_2Er_3	40.0	60.0	0.0
	B2	46.5	49.6	3.9
U2	L	28.1	61.3	10.6
	Al_2Er_3	40.0	60.0	0.0
	$AlEr_2$	33.3	66.7	0.0
	B2	32.6	49.3	18.1
U3	L	22.0	70.4	7.6
	A3-(Er)	0.0	84.7	15.3
	A2	0.0	79.2	20.8
	$AlEr_2$	33.3	66.7	0.0
U4	L	8.2	54.8	37.0
	$AlEr_2$	33.3	66.7	0.0
	A2	0.0	68.6	31.4
	B2	24.0	49.5	26.5
U5	$AlEr_2$	33.3	66.7	0.0
	A2	0.0	77.5	22.5
	A3-(Er)	0.0	84.6	15.4
	B2	21.4	50.0	28.6
D1	L	0.0	15.5	84.5
	C14	0.0	32.0	68.0
	B2	10.9	44.8	44.3
	A12-(Er,Mg)	0.0	16.8	83.2
U6	L	0.3	8.5	91.2
	C15	52.2	33.3	14.5
	A3-(Mg)	0.0	5.4	94.6
	B2	26.0	47.1	26.9
E1	L	0.2	9.2	90.6
	A3-(Mg)	0.0	6.1	93.9
	B2	24.2	47.2	28.6
	A12-(Er,Mg)	0.1	15.4	84.5
U7	L	72.1	0.1	27.8
	A1	89.2	0.0	10.8
	$L1_2$	74.4	25.0	0.6
	τ	66.7	10.0	23.3
U8	A3-(Mg)	0.0	3.7	96.3
	B2	25.9	48.3	25.8
	C15	53.7	33.3	13.0
	A12-(Er,Mg)	0.1	15.6	84.3
U9	L	42.6	0.0	57.4
	τ	66.7	10.0	23.3
	A12-(Al,Mg)	44.6	0.0	55.4
	$L1_2$	73.0	25.0	2.0
D2	L	63.8	0.0	36.2
	A1	83.4	0.0	16.6
	Al_3MG_2	61.1	0.0	38.9
	τ	66.7	10.0	23.3

(continued)

Table 2 Calculated Phase Compositions at the Invariant Equilibria in the Al-Er-Mg System (continued)

Invariant Equilibrium	Involved Phases	Composition		
		at.% Al	at.% Er	at.% Mg
D3	L	57.6	0.0	42.4
	Al_3MG_2	61.1	0.0	38.9
	A12-(Al,Mg)	51.9	0.0	48.1
	τ	66.7	10.0	23.3
U10	L	34.7	0.0	65.3
	$L1_2$	72.3	25.0	2.7
	C15	66.7	33.0	0.3
	A12-(Al,Mg)	41.4	0.0	58.6
D4	L	31.0	0.0	69.0
	A12-(Al,Mg)	39.9	0.0	60.1
	C15	66.7	33.1	0.2
	A3-(Mg)	11.6	0.0	88.4
U11	A12-(Al,Mg)	41.3	0.0	58.7
	C15	66.7	33.3	0.0
	$L1_2$	74.2	25.0	0.8
	A3-(Mg)	3.2	0.0	96.8

than those experimentally detected. Invariant equilibria involving the Al-Mg binary phases are not visible in Fig. 6 because they are very close to the Al-Mg binary subsystem.

The vertical section at 50 at.% Er is shown in Fig. 7. It is not a pseudobinary one and does not include the larger part of the B2 homogeneity region, which is mainly displaced toward Er-poorer compositions. This can be noticed in the isothermal sections of Fig. 4 and 5. Saturation surfaces delineating the B2 homogeneity region are very steep, and the aspect of the vertical sections below the liquidus curve change drastically at slighly lower or higher Er content. As a consequence, a comparison with the thermal effects detected in the experimental investigation of this section is very difficult because small composition displacements (within the experimental uncertainty) imply very large temperature variations of the detected thermal effects. Nevertheless, the DTA cooling effects shown by the samples prepared in the section have been marked in Fig. 7. Most of them may be interpreted on the basis of the computed phase equilibria.

A few more vertical sections, at 10 at.% Al, 10 at.% Er, and 10 at.% Mg, are presented in Fig. 8 to 10, respectively. The comparison between the computed equilibria and the DTA effects shown by the samples laying in the sections allowed, in some cases, the interpretation of the thermal effects. Full agreement could not be expected for many reasons. In particular, several samples could not maintain an equilibrium state during the whole thermal analysis treatment because they require long annealing times to reach the equilibrium, as underlined in part I. In some cases, moreover, the reported thermal effects correspond to equilibria close to, but not exactly included in, the section. Similar considerations may be applied to the thermal effects reported in part I pertaining to other samples not included in the sections shown here.

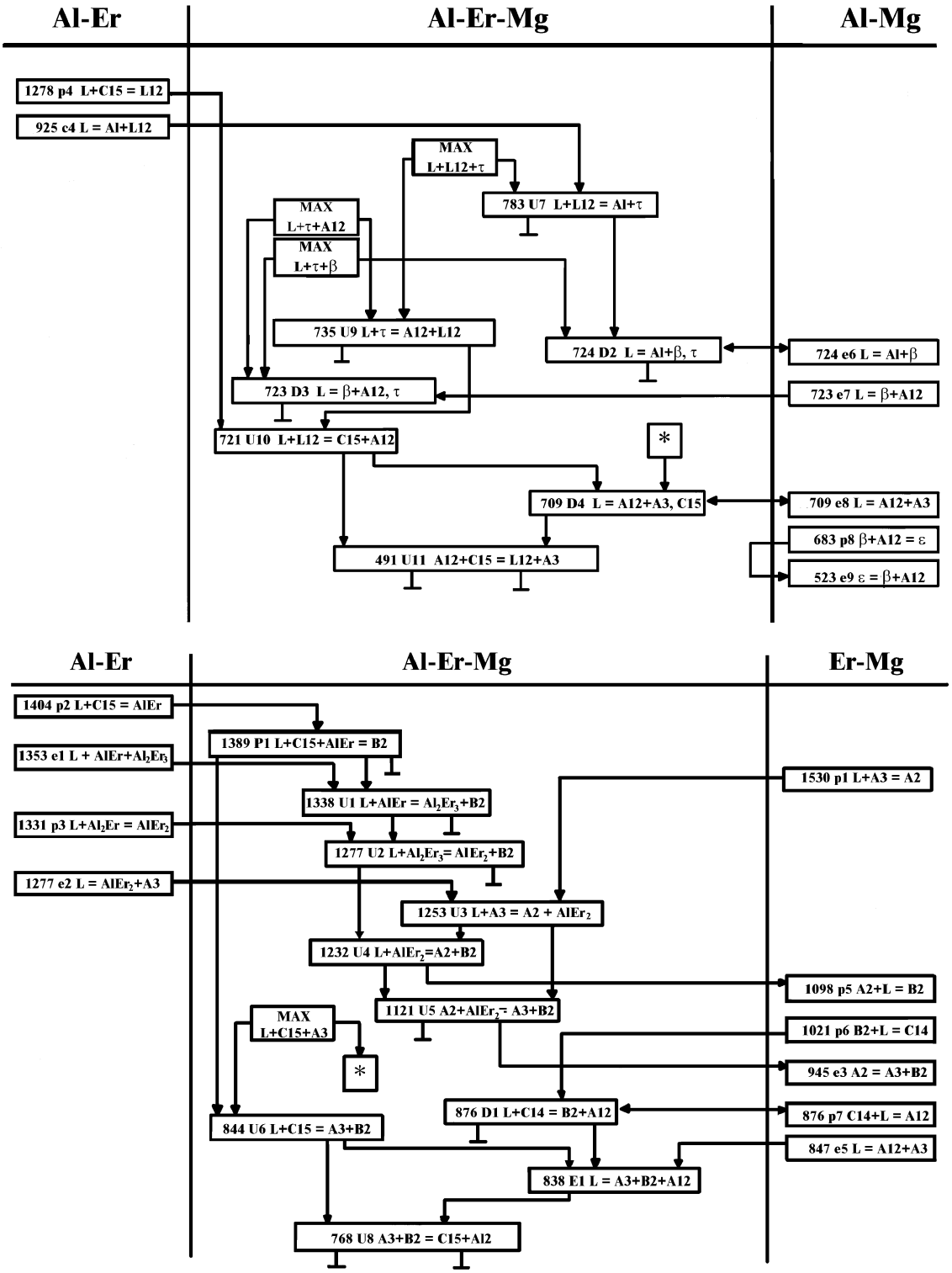

Fig. 11 The Scheil reaction scheme of the Al-Er-Mg system

The section at 10 at.% Er (Fig. 9) includes the ternary τ phase. It may be observed, in agreement with the experimental results, that its formation occurs at about 530 °C from $L1_2$ and liquid.

Finally, the sequence of the computed invariant equilibria in the system is summarized in Table 2 and shown in Fig. 11 in the form of a Scheil diagram. Most of the invariant temperatures correspond, within the experimental uncertainties, to thermal effects experimentally detected and reported in part I of this paper.

6. Conclusion

On the basis of the experimental results obtained in part I, the complete Al-Er-Mg phase diagram was thermodynamically modeled and calculated. The general good agreement between the optimized phase equilibria and the experimental information supports the reliability of the calculated phase equilibria in the whole temperature and composition range of the phase diagram.

Some uncertainty concerns the ternary equilibria of the A2 solid solution. On one hand, no clear experimental evidence of its ternary solubility was obtained, A2 being stable at temperatures higher than that of the investigated isothermal section; on the other hand, the close structural relationship with B2, which dissolves a large amount of Al, would suggest a correspondingly appreciable Al solubility in A2. In the present calculation, with no experimental evidence being available, the Al solubility in the A2 phase was not considered.

For most phases (with the exception of the B2 solid solution), the optimization required no or small ternary interaction parameters. This supports the assumption that, in many cases, a ternary-phase diagram calculated by an ideal combination of the binary ones (when these are well established and modeled) may be a quite good approximation of the real ternary-phase equilibria. The success of the modeling and the relatively small number of samples needed for the experimental investigation of the system confirm once more that a recursive approach (i.e., experiments and thermodynamic modeling) is a convenient strategy for the investigation of multicomponent and multiphase systems characterized by quite complex equilibria.

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Appendix

Functions

GHSERAL

$298.14 < T < 700$: $-7976.15 + 137.093038\,T - 24.3671976\,T\ln(T) - 0.001884662\,T^2 - 8.77664\cdot10^{-7}\,T^3 + 74092\,T^{-1}$

$700 < T < 933.47$: $-11276.24 + 223.048446\,T - 38.5844296\,T\ln(T) + 0.018531982\,T^2 - 5.764227\cdot10^{6}\,T^3 + 74092\,T^{-1}$

$933.47 < T < 2900$: $-11278.378 + 188.684153\,T - 31.748192\,T\ln(T) - 1.230524\cdot10^{28}\,T^{-9}$

GLIQAL

$298.14 < T < 700$: $+3028.879 + 125.251171\,T - 24.3671976\,T\ln(T) - 0.001884662\,T^2 + 74092\,T^{-1} - 8.77664\cdot10^{-7}\,T^3 + 7.934\cdot10^{-20}\,T^7$

$700 < T < 933.47$: $-271.21 + 211.206579\,T - 38.5844296\,T\ln(T) + 0.018531982\,T^2 + 74092\,T^{-1} - 5.764227\cdot10^{-6}\,T^3 + 7.934\cdot10^{-20}\,T^7$

$933.47 < T < 2900$: $-795.996 + 177.430178\,T - 31.748192\,T\ln(T)$

GHSERER

$298.14 < T < 1802$: $-8489.136 + 116.698964\,T - 28.3846744\,T\ln(T) + 9.95792\cdot10^{-4}\,T^2 - 9.52557\cdot19^{-7}\,T^3 + 9581\,T^{-1}$

$1802 < T < 3200$: $-445688.206 + 2233.10212\,T - 298.135131\,T\ln(T) + 0.065950553\,T^2 - 3.041405\cdot10^{-6}\,T^3 + 1.23973199\cdot10^{8}\,T^{1}$

GLIQER

$298.14 < T < 500$: $+19382.102 - 10.241846\,T + GHSERER$

$500 < T < 1802$: $+17912.678 + 0.355564\,T - 12.0761776\,T\ln(T) - 0.014414687\,T^2 + 1.316517\cdot10^{-6}\,T^3 - 528122\,T^{-1}$

$1802 < T < 3200$: $+747.131 + 187.623024\,T - 38.702\,T\ln(T)$

GHSERMG

$298.14 < T < 923$: $-8367.34 + 143.677875\,T - 26.1849782\,T\ln(T) + 4.858\cdot10^{-4}\,T^2 - 1.393669\cdot10^{-6}\,T^3 + 78950\,T^{-1}$

$923 < T < 3000$: $-14130.185 + 204.718543\,T - 34.3088\,T\ln(T) + 1.038192\cdot10^{28}\,T^{-9}$

GLIQMG

$298.14 < T < 923$: $+8202.243 - 8.83693\,T + GHSERMG - 8.0176\cdot10^{-20}\,T^7$

$923 < T < 3000$: $-5439.869 + 195.324057\,T - 34.3088\,T\ln(T)$

Phases

Liquid Model: (Al,Er,Mg)

$G(\text{Liquid,Al;0}) - H^{298}(\text{A1;Al;0}) = +GLIQAL$

$G(\text{Liquid,Er;0}) - H^{298}(\text{A3;Er;0}) = +GLIQER$

$G(\text{Liquid,Mg;0}) - H^{298}(\text{A3;Mg;0}) = +GLIQMG$

$L(\text{Liquid,Al,Er;0}) = -\underline{176486} + \underline{55.6852}\,T$

$L(\text{Liquid,Al,Er;1}) = -\underline{36685.5} + \underline{23.4492}\,T$

L(Liquid,Al,Er;2) = +34349.1 − 8.23519 T
L(Liquid,Al,Mg;0) = −12000 + 8.566 T
L(Liquid,Al,Mg;1) = +1894 − 3 T
L(Liquid,Al,Mg;2) = 2000
L(Liquid,Er,Mg;0) = −40308.8 + 15.7946 T
L(Liquid,Er,Mg;1) = −38228 − 20.0235 T
L(Liquid,Al,Er,Mg;0) = +9.75996 T

A2 Model: (Al,Er,Mg)
This is also the disordered part of B2

G(A2,Al;0) − H^{298}(A1;Al;0) = +10083 − 4.813 T + *GHSERAL*
G(A2,Er;0) − H^{298}(A3;Er;0) = +*GHSERER* + 4600 − 0.3 R T
G(A2,Mg;0) − H^{298}(A3;Mg;0) = +3100 − 2.1 T + *GHSERMG*
L(A2,Er,Mg;0) = −40411 + 20.557 T
L(A2,Er,Mg;1) = 1969

B2 Model: $(Al,Er,Mg)_{0.5}(Al,Er,Mg)_{0.5}$
Disordered contribution from A2

G(B2,Al:Al;0) − H^{298}(A1;Al;0) = 0.0
G(B2,Er:Er;0) − H^{298}(A3;Er;0) = 0.0
G(B2,Mg:Mg;0) − H^{298}(A3;Mg;0) = 0.0

G(B2,Al:Er;0) − 0.5 H^{298}(A1;Al;0) − 0.5 H^{298}(A3;Er;0) = −107332 + 26.2958 T
G(B2,Er:Al;0) − 0.5 H^{298}(A1;Al;0) − 0.5 H^{298}(A3;Er;0) = −107332 + 26.2958 T

G(B2,A1:Mg;0) − 0.5 H^{298}(A1;Al;0) − 0.5 H^{298}(A3;Mg;0) = 0.0
G(B2,Mg:Al;0) − 0.5 H^{298}(A1;Al;0) − 0.5 H^{298}(A3;Mg;0) = 0.0

G(B2,Er:Mg;0) − 0.5 H^{298}(A3;Er;0) − 0.5 H^{298}(A3;Mg;0) = −28325 + 29.312 T
G(B2,Mg:Er;0) − 0.5 H^{298}(A3;Er;0) − 0.5 H^{298}(A3;Mg;0) = −28325 + 29.312 T
L(B2,Er:Mg:Er;0) = 25443.4
L(B2,Er:Er:Mg;0) = 25443.4
L(B2,Mg:Er:Mg;0) = 3320.88
L(B2,Er:Mg:Mg;0) = 3320.88

L(B2,Er:Al:Mg;0) = −54110 + 36.5785 T
L(B2,Al:Mg:Er;0) = −54110 + 36.5785 T

A1 Model: (Al,Er,Mg)
This is also the disordered part of $L1_2$

G(A1,Al;0) − H^{298}(A1;Al;0) = +*GHSERAL*
G(A1,Er;0) − H^{298}(A3;Er;0) = +1950 − 0.1 R T + *GHSERER*
G(A1,Mg;0) − H^{298}(A3;Mg;0) = +2600 − 0.9 T + *GHSERMG*

L(A1,Al,Mg;0) = +4971 − 3.5 T
L(A1,Al,Mg;1) = +900 + 0.423 T
L(A1,Al,Mg;2) = 950

L(A1,Er,Mg;0) = +29097

$L1_2$ Model: $(Al,Er,Mg)_{0.75}(Al,Er,Mg)_{0.25}$
Disordered contribution from A1

G(L12,Al:Al;0) − H^{298}(A1;Al;0) = 0.0
G(L12,Er:Er;0) − H^{298}(A3;Er;0) = 0.0
G(L12,Mg:Mg;0) − H^{298}(A3;Mg;0) = 0.0

G(L12,Al:Er;0) − 0.75 H^{298}(A1;Al;0) − 0.25 H^{298}(A3,Er;0) = +*GAL3ER*
G(L12,Er:Al;0) − 0.25 H^{298}(A1;Al;0) − 0.75 H^{298}(A3,Er;0) = +*GALER*3
L(L12,Al:Al,Er;0) = +*L04ALER*
L(L12,Al:Al,Er;1) = +*L14ALER*

L(L12,Al,Er:Al;0) = −1.5 *GALER*3 + 1.5 *GAL2ER*2 + 1.5 GAL3*ER* + 3 *L04ALER*
L(L12,Al,Er:Al;1) = +0.5 *GALER*3 − 1.5 *GAL2ER*2 + 1.5 GAL3*ER* + 3 *L14ALER*
L(L12,Er:Al,Er;0) = +*L04ALER*
L(L12,Er:Al,Er;1) = +*L14ALER*
L(L12,Al,Er:Er;0) = +1.5 *GALER*3 + 1.5 *GAL2ER*2 − 1.5 GAL3*ER* + 3 *L04ALER*
L(L12,Al,Er:Er;1) = −1.5 *GALER*3 + 1.5 *GAL2ER*2 − 0.5 GAL3*ER* + 3 *L14ALER*

G(L12,Al:Mg;0) − 0.75 H^{298}(A1,Al;0) − 0.25 H^{298}(A3,Mg;0) = 0.0
G(L12,Mg:Al;0) − 0.25 H^{298}(A1,Al;0) − 0.75 H^{298}(A3,Mg;0) = 0.0
G(L12,Er:Mg;0) − 0.75 H^{298}(A3,Er;0) − 0.25 H^{298}(A3,Mg;0) = 0.0
G(L12,Mg:Er;0) − 0.25 H^{298}(A3,Er;0) − 0.75 H^{298}(A3,Mg;0) = 0.0

with: *GAL3ER* = +3 (−$\underline{25086}$ + $\underline{6.1647}$ T)
*GALER*3 = 0.0
*GAL2ER*2 = 0.0
L04ALER = 0.0
L14ALER = 0.0

A3 Model: (Al,Er,Mg)

G(A3,Al;0) − H^{298}(A1,Al;0) = +5481 − 1.8 T + *GHSERAL*
G(A3,Er;0) − H^{298}(A3,Er;0) = +*GHSERER*
G(A3,Mg;0) − H^{298}(A3,Mg;0) = +*GHSERMG*

L(A3,Al,Mg;0) = +1950 − 2 T
L(A3,Al,Mg;1) = +1480 − 2.08 T
L(A3,Al,Mg;2) = 3500

L(A3,Er,Mg;0) = −$\underline{20671.3}$ + $\underline{7.23489}$ T
L(A3,Er,Mg;1) = $\underline{10794.9}$

A12 Model: $(Er,Mg)_5(Al,Mg)_{12}(Al,Mg)_{12}$

G(A12,Mg:Mg:Mg;0) − 29 H^{298}(A3,Mg;0) = +133469.6 − 87.319 T + 29 *GHSERMG*
G(A12,Er:Al:Al;0) − 24 H^{298}(A1,Al;0) − 5 H^{298}(A3,Er;0) = −$\underline{412594.6}$ + $\underline{206.3}$ T + 5 *GHSERER* + 24 *GHSERAL*
G(A12,Mg:Al:Al;0) − 24 H^{298}(A1,Al;0) − 5 H^{298}(A3,Mg;0) = +97875 − 101.5 T + 5 *GHSERMG* + 24 *GHSERAL*
G(A12,Mg:Mg:Al;0) − 12 H^{298}(A1,Al;0) − 17 H^{298}(A3,Mg;0) = −52780 − 50.75 T + 17 *GHSERMG* + 12 *GHSERAL*
G(A12,Mg:Al:Mg;0) − 12 H^{298}(A1,Al;0) − 17 H^{298}(A3,Mg;0) = +284124.6 − 138.069 T + 17 *GHSERMG* + 12 *GHSERAL*
L(A12,Mg:Al:Al,Mg;0) = +113100 − 14.5 T
L(A12,Mg:Mg:Al,Mg;0) = +113100 − 14.5 T
G(A12,Er:Mg:Mg;0) − 5 H^{298}(A3,Er;0) − 24 H^{298}(A3,Mg;0) = −$\underline{377000}$ + $\underline{201.5}$ T + 5 *GHSERER* + 24 *GHSERMG*
G(A12,Er:Mg:Al;0) − 12 H^{298}(A1,Al;0) − 5 H^{298}(A3,Er;0) = −12 H^{298}(A3,Mg;0) = −$\underline{563249.6}$ + $\underline{281.6}$ T + 5 *GHSERER* + 12 *GHSERMG* + 12 *GHSERAL*
G(A12,Er:Al:Mg;0) − 12 H^{298}(A1,Al;0) − 5 H^{298}(A3,Er;0) = −12 H^{298}(A3,Mg;0) = −$\underline{226345}$ + $\underline{113.2}$ T + 5 *GHSERER* + 12 *GHSERMG* + 12 *GHSERAL*

C14 Model: $(Er,Mg)_2(Er,Mg)_1$

G(C14,Er:Er;0) − 3 H^{298}(A3,Er;0) = +15000 + 3 *GHSERER*
G(C14,Mg:Mg;0) − 3 H^{298}(A3,Mg;0) = +15000 + 3 *GHSERMG*

G(C14,Mg:Er;0) − H^{298}(A3,Er;0) − 2 H^{298}(A3,Mg;0) = +*GC14ERMG* + 2 *GHSERMG* + *GHSERER*
G(C14,Er:Mg;0) − 2 H^{298}(A3,Er;0) − H^{298}(A3,Mg;0) = −*GC14ERMG* + 30000 + 2 *GHSERER* + *GHSERMG*

with: *GC14ERMG* = −$\underline{64500}$ + $\underline{36.35}$ T

C15 Model: $(Al,Er,Mg)_2(Al,Er,Mg)_1$

G(C15,Al:Al;0) − 3 H^{298}(A1,Al;0) = +15000 + 3 *GHSERAL*
G(C15,Er:Er;0) − 3 H^{298}(A3,Er;0) = +15000 + 3 *GHSERER*
G(C15,Mg:Mg;0) − 3 H^{298}(A3,Mg;0) = +15000 + 3 *GHSERMG*

G(C15,Al:Er;0) − 2 H^{298}(A1,Al;0) − H^{298}(A3,Er;0) = +*GC15ALER* + 2 *GHSERAL* + *GHSERER*
G(C15,Er:Al;0) − H^{298}(A1,Al;0) − 2 H^{298}(A3,Er;0) = −*GC15ALER* + 30000 + 2 *GHSERER* + *GHSERAL*

G(C15,Al:Mg;0) − 2 H^{298}(A1,Al;0) − H^{298}(A3,Mg;0) = +15000 + 2 *GHSERAL* + *GHSERMG*
G(C15,Mg:Al;0) − H^{298}(A1,Al;0) − 2 H^{298}(A3,Mg;0) = +15000 + 2 *GHSERMG* + *GHSERAL*
G(C15,Mg:Er;0) − H^{298}(A3,Er;0) − 2 H^{298}(A3,Mg;0) = +*GC15ERMG* + 2 *GHSERMG* + *GHSERER*
G(C15,Er:Mg;0) − 2 H^{298}(A3,Er;0) − H^{298}(A3,Mg;0) = −*GC15ERMG* + 30000 + 2 *GHSERER* + *GHSERMG*

with: *GC15ALER* = −165000 + 40.338 *T*
GC15ERMG = +*GC14ERMG* + 14379 − 1.5653 *T*
GC14ERMG = −64500 + 36.35 *T*

AlEr Model: $(Al,Mg)_{0.5}(Er)_{0.5}$

G(AlEr,Al:Er;0) − 0.5 H^{298}(A1,Al;0) − 0.5 H^{298}(A3,Er;0) = −50000 + 11.992178 *T* + 0.5 *GHSERAL* + 0.5 *GHSERER*
G(AlEr,Mg:Er;0) − 0.5 H^{298}(A3,Er;0) − 0.5 H^{298}(A3,Mg;0) = +0.5 *GHSERMG* + 0.5 *GHSERER*

Al_2Er_3 Model: $(Al)_{0.4}(Er)_{0.6}$

G(Al_2Er_3,Al:Er;0) − 0.4 H^{298}(A1,Al;0) − 0.6 H^{298}(A3,Er;0) = −45000 + 10.8726018 *T* + 0.4 *GHSERAL* + 0.6 *GHSERER*

$AlEr_2$ Model: $(Al)_{0.333}(Er)_{0.667}$

G($AlEr_2$,Al:Er;0) − 0.33333 H^{298}(A1,Al;0) − 0.66667 H^{298}(A3,Er;0) = −39000 + 9.2227795 *T* + 0.33333 *GHSERAL* + 0.66667 *GHSERER*

β Model: $(Mg)_{89}(Al)_{140}$

G(β,Mg:Al;0) − 140 H^{298}(A1,Al;0) − 89 H^{298}(A3,Mg;0) = −246175 − 675.55 *T* + 89 *GHSERMG* + 140 *GHSERAL*

ε Model: $(Mg)_{23}(Al)_{30}$

G(ε,Mg:Al;0) − 30 H^{298}(A1,Al;0) − 23 H^{298}(A3,Mg;0) = −52565.4 − 173.1775 *T* + 23 *GHSERMG* + 30 *GHSERAL*

τ Model: $(Al)_{0.667}(Er)_{0.1}(Mg)_{0.233}$

G(τ,Al:Er:Mg;0) − 0.66667 H^{298}(A1,Al;0) − 0.1 H^{298}(A3,Er;0) − 0.23333 H^{298}(A3,Mg;0) = −18000 + 1.8899 *T* + 0.66667 *GHSERAL* + 0.1 *GHSERER* + 0.23333 *GHSERMG*

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